

SHISHIR SINGH

Computer Science undergraduate with hands-on experience in Linux systems programming, Bash automation, Computer Networks.

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WORK EXPERIENCES

Systems Engineer (Embedded) | Internship

May 2026 - Aug 2026

Ciena Corporation

On-site • Gurugram

- Automated network hardware monitoring by deploying **Bash scripts** across 25 switches, enabling instant port-level connectivity validation and reducing diagnostic reliance on the hardware engineering team.
- Executed Telnet-based communication workflows** for efficient remote device management and automated test result collection.
- Enhanced process communication** and synchronization using **Linux IPC mechanisms**, including pipes, shared memory, sockets, and message queues.
- Troubleshooted and debugged Linux-based software workflows** using shell utilities, logs, and networking tools to ensure operational integrity.
- Collaborated with engineers** on telecom networking systems, acquiring expertise in access networks, protocol-driven software development, and embedded **Linux** environments.

Technologies / Skills Used : Linux, Bash, Telnet, IPC mechanisms, cron, SSH

PROJECTS

Linux IPC Monitoring Framework | 🐙 Github

Developer

- Designed and developed a Linux system-programming project** in C utilizing pipes, shared memory, and POSIX message queues for IPC.
- Implemented producer-consumer architecture** with concurrent processes and process synchronization mechanisms.
- Optimized inter-process data exchange** to achieve low-latency communication between independent processes.
- Built logging, fault-handling, and monitoring modules to improve reliability and debugging efficiency.
- Validated system stability through stress testing and concurrent workload simulation.

Technologies / Tools Used : C, POSIX, pipes, shared memory

TCP Client-Server File Transfer System

Developer

- Developed a client-server application** in Python using TCP sockets for reliable file transfer across networked systems.
- Implemented connection establishment, packet validation, timeout handling, retransmission logic, and error recovery mechanisms.
- Designed networking workflows following **TCP/IP** communication principles and socket programming concepts.
- Analyzed network traffic using **Wireshark** and **tcpdump** to debug protocol behavior and optimize communication reliability.

Technologies / Tools Used : Python, TCP sockets, Wireshark, tcpdump

SKILLS

Programming Languages :	C++, Python, Bash, SQL
Databases :	PostgreSQL, MySQL
Tools & Platforms :	Linux CLI, AWS EC2, Basic CC concepts, Git, GitHub, Mobaxterm
Soft Skills :	Communication, Adaptability, Teamwork, Critical Thinking
Languages :	English, Hindi

EDUCATION

Vellore Institute of Technology | India

2023 - 2027

B.Tech. Computer Science and Engineering spec. E-Commerce Technology

CGPA : 8.00

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Computer Networks

AWARDS & ACHIEVEMENTS

- Accepted 300+** Data Structures and Algorithms problems on Codeforces and competitive programming platforms; achieved **Codeforces Pupil rating** of 1214.
- Successfully cleared **GenW.AI Level 1** at Deloitte Hacksplosion 2026 Hackathon.[.link](#)

EXTRACURRICULAR

- Running.
- Solving Ad-hoc problem on competitive programming platforms like codeforces and codechef.